

Human-Like Chess: Report

Motivation:

Modern chess engines can be adjusted to play at a wide range of Elo ratings through techniques such as limiting search depth, restricting computational resources, or deliberately selecting suboptimal moves. While these approaches can successfully match the overall strength of human players at a given rating level, the resulting play often remains distinctly non-human. The mistakes made by weakened engines frequently differ from those made by humans of comparable skill, motivating the development of models that better emulate human decision-making processes.

Core Hypothesis:

Rather than weakening a superhuman engine, human-like chess play can be more accurately reproduced by training models to predict the moves selected by players of a given Elo rating. Such models should naturally learn the strengths, weaknesses, and characteristic mistakes associated with human play at different skill levels.

Dataset:

The first requirement of this project was a large collection of chess games played by humans across a wide range of skill levels. To obtain this data, Chess.com's public API was used to retrieve the first 10,000 players from ten countries with large and active chess communities: the United States, India, Brazil, Indonesia, the Philippines, Great Britain, Canada, Germany, France, and Turkey.

Player statistics were then retrieved for each account. This project focused exclusively on Rapid time controls, as Rapid ratings generally provide a more accurate measure of chess skill than Blitz or Bullet ratings, where time pressure, intuition, and opening preparation have a greater influence on move selection. Restricting the dataset to Rapid games also reduced the number of suboptimal moves attributable primarily to time constraints rather than player ability.

A central objective of this project was to develop models capable of emulating human players at different skill levels. As an initial test of this approach, separate datasets were constructed for players rated approximately 1000 Elo and 2000 Elo, representing novice-level and expert-level play, respectively.

To construct these datasets, several filtering criteria were applied. First, only players whose most recent Rapid rating fell within ± 100 Elo of the target rating were considered. Second,

players were required to have completed at least 50 Rapid games and maintain a rating deviation below 100 points. These filters increased confidence that the recorded rating accurately reflected each player's true playing strength.

Following filtering, 500 players from each Elo group were selected to continue through the data engineering pipeline. To maintain manageable dataset sizes while ensuring the data reflected current playing strength, only games played within the previous 24 months were considered. These constraints were necessary to keep training times and hardware requirements within practical limits, as players in the dataset typically had extensive game histories. For example, players near the 1000 Elo range had a median of approximately 850 Rapid games played.

The Chess.com games API was queried for each month within the 24-month observation window for every selected player. Retrieved games were filtered to include only rated Rapid games played under standard chess rules in which both participants fell within ± 100 Elo of the target rating. To prevent a small number of highly active players from disproportionately influencing the dataset, the number of games retained per player was capped at the median game count for the corresponding Elo group.

Feature Engineering:

Each game in the dataset was transformed into a collection of training samples by extracting the board state after every half-move. For each sample, the current position, the subsequent move played, the average Elo rating of the two players, and the Chess.com game review URL (used as a unique game identifier) were recorded. This transformation expanded the datasets to approximately 10 million and 14 million training samples for the 1000 Elo and 2000 Elo groups, respectively.

The board position associated with each sample was then encoded as a sequence of tokens representing the complete game state, including the location of all pieces, the side to move, and the castling rights available to each player.

Finally, all legal UCI move representations (e.g., e2e4) were mapped to unique integer identifiers. The move played by the human participant was then encoded using this mapping and served as the target variable during model training. The encoded game state and associated numeric representation of the move would be all that was sent into the model for training and prediction.

Train/Test Split:

A random train/test split at the training-sample level would introduce the possibility of data leakage, as positions from the same game could appear in both the training and test sets.

Because consecutive positions within a game are highly correlated, the model could effectively gain indirect knowledge of test-set positions through related positions seen during training.

To prevent this, the dataset was split at the game level rather than the sample level. All positions derived from a given game were assigned exclusively to either the training set or the test set. This ensured that the model was evaluated only on games it had never encountered during training, providing a more reliable measure of its ability to generalize to unseen play.

Model Details:

A transformer-based architecture was selected to model human chess decision making. Unlike traditional machine learning approaches, which often require extensive manual feature engineering, transformers can learn relationships directly from tokenized board representations. This makes them well suited to chess, where the quality of a move depends on complex interactions between pieces distributed across the board.

Each board position was represented as a sequence of tokens encoding the locations of all pieces, the side to move, and available castling rights. These tokens were converted into learned embeddings and combined with positional embeddings to preserve information about their location within the input sequence.

The embedded positions were processed by a four-layer transformer encoder with eight attention heads per layer and an embedding dimension of 128. This configuration was chosen as a balance between representational capacity and computational efficiency. The model was sufficiently expressive to learn relationships between pieces and board structures while remaining practical to train on datasets containing tens of millions of training samples.

The output of the transformer encoder was aggregated through mean pooling to produce a single representation of the board position. A final linear classification layer projected this representation to a probability distribution over all encoded UCI moves, allowing the model to predict the move most likely to be selected by a human player in the given position.

Model Training:

Move prediction was formulated as a multiclass classification problem in which the model was tasked with predicting the human move associated with a given board position. The output layer therefore contained one neuron for each encoded UCI move present in the dataset.

The model was trained using cross-entropy loss and optimized using AdamW with a learning rate of 1×10^{-4} . AdamW was selected due to its strong empirical performance on transformer-based architectures and its ability to provide stable optimization on large datasets.

Each board position was represented as a sequence of 69 tokens drawn from a vocabulary of 23 unique symbols encoding piece identities and game-state information. Models were trained for 10 epochs, balancing training time against model convergence.

Training was performed using PyTorch on an NVIDIA GeForce RTX 4060 Laptop GPU. Under this configuration, each model required approximately 12 hours to train.

Evaluation Methodology:

Model performance was evaluated using Top-1 and Top-5 accuracy on the held-out test set. Top-1 accuracy measures the percentage of positions for which the model's highest-probability prediction exactly matched the move played by the human player. While informative, this metric can be overly strict in chess, as many positions admit multiple reasonable candidate moves.

To provide a broader measure of predictive performance, Top-5 accuracy was also calculated. A prediction was considered correct if the move played by the human appeared among the model's five highest-probability move predictions. This metric captures the model's ability to identify plausible human choices even when its highest-ranked prediction differs from the move ultimately selected.

Results:

Model Elo Rating	Top-1 Accuracy (%)	Top-5 Accuracy (%)
1000	36.64	73.63
2000	36.61	73.77

These results indicate that the models were frequently able to identify the move chosen by the human player and, in the majority of positions, ranked the human move among their most likely predictions. Achieving a Top-1 accuracy exceeding 36% for both models is notable given that many chess positions admit several moves of comparable quality, any of which may be selected by a human player. The Top-5 accuracy of nearly 75% is particularly significant, suggesting that the models were often able to identify a small set of plausible candidate moves even when the human-selected move was not ranked first. This behavior is consistent with the objective of modeling human decision-making, where multiple reasonable alternatives may exist in each position.

Observations:

When playing against the 1000 Elo model, its opening play generally followed common opening principles and frequently mirrored popular opening lines observed in human games. However, as games progressed into the middlegame, the model began exhibiting characteristics commonly associated with novice players, including unfavorable material exchanges, missed tactical opportunities, overlooked hanging pieces, and an apparent tendency to prioritize immediate captures over longer-term positional considerations. Overall, the model's play was consistent with expectations for a player near the 1000 Elo level.

In contrast, the 2000 Elo model demonstrated substantially stronger play throughout all phases of the game. Its opening choices more closely matched established opening theory, and it maintained a higher level of tactical and positional accuracy through the middlegame and endgame. Compared to the 1000 Elo model, its decisions appeared more purposeful and strategically coherent, closely resembling the play style expected from an experienced club or tournament player.

Limitations:



Figure 1: Human player playing as White against model trained on 2000 Elo chess games. Model misses the hanging rook due to this being unnatural development at 2000 Elo level, and having no explicit positional evaluation.

Both models exhibited difficulty adapting to positions that were uncommon within the training data. For example, when presented with an opening position in which an opponent's rook was intentionally placed en prise to an undeveloped bishop, the 2000 Elo model often failed to capitalize on the tactical opportunity. Instead, it frequently continued with standard opening moves such as piece development and central control, despite the capture being clearly advantageous according to traditional chess engines such as Stockfish.

A similar limitation was observed in simplified endgame positions. When only a small number of pieces remained on the board, both models often struggled to convert winning positions into checkmate. While the models generally selected moves that appeared

reasonable, they lacked the consistency and precision required to execute many fundamental endgame techniques.

These behaviors likely stem from the fact that the models were trained solely to predict human moves rather than to optimize a positional evaluation function. Unlike traditional chess engines, which explicitly evaluate board positions and search for moves that maximize positional advantage, the transformer learned only from observed human decisions. As a result, it performed best in positions that closely resembled those encountered during training and was less reliable in unusual tactical situations, or in endgame scenarios.

Future Work:

There are two primary avenues for future improvement. First, training the models on a larger and more diverse collection of games would expose them to a wider variety of chess positions, particularly those that occur infrequently in normal play. This would likely improve the model's ability to respond appropriately in unusual tactical and endgame situations. However, because such positions represent only a small fraction of all positions encountered in human games, the benefits of additional training data alone are expected to be incremental.

Additionally, it would be valuable to apply the same methodology to a broader range of Elo ratings. The present work focused on 1000- and 2000-rated players as representative examples of novice- and expert-level play, respectively. Future investigations could examine whether models trained on more incremental rating groups, such as 1200, 1400, 1600, and 1800 Elo exhibit distinguishable differences in playing style and decision-making patterns. This would help determine whether the model is capable of capturing the subtle differences in play that separate nearby Elo ranges.

A potentially more significant improvement to the model's play would be the incorporation of positional evaluation into the training process. The current models learn exclusively from human move selection and therefore have no explicit understanding of whether a position is advantageous or disadvantageous. Integrating information from a chess engine evaluation function could enable the model to balance human-like move preferences with objective positional strength. Such an approach may improve performance in rare tactical situations, help the model identify opportunities that are uncommon in the training data, and increase its ability to convert winning positions into checkmate while preserving its human-like playing style.

Conclusion:

This project investigated whether human chess play could be modeled directly from historical game data rather than by weakening a traditional chess engine. By training transformer-based neural networks on nearly 200,000 human games for each target rating group, the resulting models learned to reproduce many of the characteristics associated with players at different skill levels.

A clear distinction was observed between the model trained on games from approximately 1000-level Elo players and the model trained on games from approximately 2000-level Elo players. The 1000 Elo model exhibited many of the strengths and weaknesses commonly associated with novice players, including tactical oversights and questionable material exchanges, while the 2000 Elo model demonstrated stronger opening knowledge, improved tactical awareness, and more consistent play throughout all phases of the game.

Although both models struggled in unusual tactical positions and simplified endgames, the results suggest that training directly on human move selections is a viable approach to modeling human chess behaviour. More broadly, this work demonstrates that machine learning models can capture not only differences in playing strength, but also the characteristic decision-making patterns that distinguish players at different skill levels.